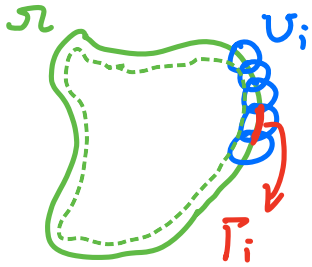
2025.12.10 + 2025.12.15 + 2025.12.17 + 2025.12.24

Application of partition of unity



compute $\int_{\partial\Omega} f(x)\,ds$

$\Gamma_i = \{x \in U_i \cap \partial\Omega : x_n = g_i(x_1, \cdots, x_{n-1})\}$ (参数化)

$$\int_{\partial\Omega} f(x)\,ds = \int_{\partial\Omega} f(x) \sum_{i=1}^{k} \varphi_i(x)\,ds = \sum_{i=1}^{k} \int_{\partial\Omega} f(x)\varphi_i(x)\,dx$$

$$= \sum_{i=1}^{k} \int_{\Gamma_i} f(x)\varphi_i(x)\,dx =$$

$$\sum_{i=1}^{k} \int_{\mathbb{R}^{n-1}} (f\varphi_i)(\underbrace{x_1, \cdots, x_{n-1}}_{x'}, g(\underbrace{x_1 \cdots x_{n-1}}_{x'})) \sqrt{1+|\nabla_x g|^2}\,dx_1 \cdots dx_{n-1}$$

Density Thm

If $1 \le p < \infty$, $k \ge 0 \in \mathbb{N}$, then $C^\infty(\Omega) \cap W^{k,p}(\Omega)$ is dense in $W^{k,p}(\Omega)$.

($k=0$, $W^{0,p}(\Omega) = L^p(\Omega)$, $C^\infty(\Omega) \cap L^p(\Omega)$ is dense in $L^p(\Omega)$)

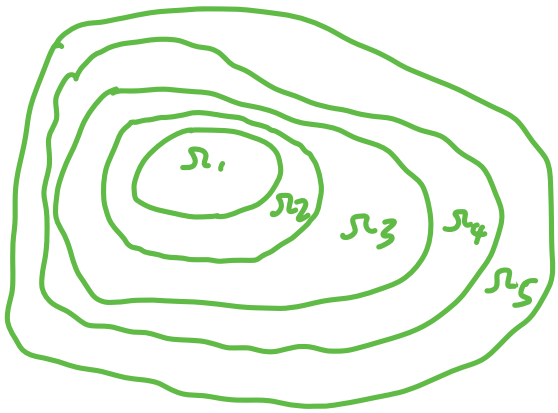
pf: $k \ge 1$, W.T.S.: $\forall u \in W^{k,p}(\Omega)$, $\forall \varepsilon > 0$, $\exists v \in C^\infty(\Omega) \cap W^{k,p}(\Omega)$ s.t.

$\|u - v\|_{W^{k,p}(\Omega)} < \varepsilon$.

Take large $R > 0$ s.t. $B_R(0) \cap \Omega \neq \emptyset$. Then let

$\Omega_j = \{x \in \Omega : x \in B_{R+j}(0), \operatorname{dist}(x, \partial\Omega) > \frac{1}{j}\}$, $\Omega_j \subset\subset \Omega_{j+1} \subset\subset \cdots \subset\subset \Omega$

$\bigcup_{j=1}^{\infty} \Omega_j = \Omega$



Let $U_j = \Omega_{j+1} \setminus \overline{\Omega}_{j-1}$, $j \ge 0$, $\Omega_0 = \Omega_{-1} = \emptyset$

$U_0 = \Omega_1$, $U_1 = \Omega_2$, $U_2 = \Omega_3 \setminus \overline{\Omega}_1$, $U_3 = \Omega_4 \setminus \overline{\Omega}_2$

$U_4 = \Omega_5 \setminus \overline{\Omega}_3$

$\Rightarrow \{U_j\}_{j=0}^{\infty}$ satisfy the condition of partition of unity.

$\Rightarrow \exists \{\phi_j\}_{j=1}^{\infty}$, $\sum_{j=1}^{\infty} \phi_j(x) = 1$, $0 \le \phi_j \le 1$, $\phi_j \in C_0^\infty(U_j)$, $\operatorname{supp} \phi_j \subset U_j$

Since $u \in W^{k,p}(\Omega) \Rightarrow \phi_j u \in W^{k,p}(\Omega)$. Consider $(\phi_j u)_n = \int_\Omega j_n(x-y)(\phi_j u)(y)\,dy$

By Lemma 5, $\partial^\alpha(\phi_j u)_h = (\partial^\alpha \phi_j u)_h$ $\forall x \in \Omega_h$ & $(x \in \Omega, \operatorname{supp}(\phi_j u) \subset\subset \Omega)$

By Lemma 4, $[\partial^\alpha(\phi_j u)]_h \xrightarrow{h\to 0} \partial^\alpha(\phi_j u)$ in $L^p(\Omega)$ $\forall \alpha, |\alpha| \le k$.

$\partial^\alpha(\phi_j u)$

$\Rightarrow (\phi_j u)_h \to \phi_j u$ in $W^{k,p}(\Omega)$ as $h \to 0$.

$\forall \varepsilon > 0$, for each $j = 0, 1, \cdots$, take $h_j > 0$ small s.t. $\|(\phi_j u)_{h_j} - \phi_j u\|_{W^{k,p}(\Omega)} < \frac{\varepsilon}{2^j}$

Let $v(x) = \sum_{j=0}^{\infty} (\phi_j u)_{h_j}(x)$. $\operatorname{supp}(\phi_j u)_{h_j} \subset\subset U_j$, $\forall$ subdomain $\Omega' \subset\subset \Omega$, Ω' intersect only finitely many U_j

$\Rightarrow v(x)$ is well-defined on Ω' and $v \in C^\infty(\Omega') \Rightarrow v \in C^\infty(\Omega)$

$\|u - v\|_{W^{k,p}} = \|\sum_{j=0}^{\infty} \phi_j u - \sum_{j=0}^{\infty} (\phi_j u)_{h_j}\|_{W^{k,p}} \le \sum_{j=0}^{\infty} \|\phi_j u - (\phi_j u)_{h_j}\| < \sum_{j=0}^{\infty} \frac{\varepsilon}{2^{j+1}} = \varepsilon$.

$v = v - u + u \in W^{k,p}(\Omega)$

Remark: If Ω is bdd, $\partial\Omega \in C^1$, then $C^\infty(\bar\Omega)$ is dense in $W^{k,p}(\Omega)$.

($k=0$, $C^\infty(\bar\Omega)$ is dense in $L^p(\Omega)$, no need $\partial\Omega \in C^1$)

Sobolev inequalities.

$W^{1,p}(\Omega) = \{u \in L^p(\Omega) : \text{weak } \nabla u \in L^p(\Omega)\}$

$W_0^{1,p}(\Omega) =$ closure of $C_c^\infty(\Omega)$ in $W^{1,p}(\Omega)$

Inequalities: $\Omega \subseteq \mathbb{R}^n$, $n \ge 2$ open, connected $C_1(n,p) = \frac{p(n-1)}{\sqrt{n}(n-p)} \to \infty$ $(p \to n^-)$

(i) If $1 \le p < n$ and $p^* = \frac{np}{n-p}$, then $\|u\|_{L^{p^*}} \le C_{n,p} \|\nabla u\|_{L^p}$, $\forall u \in W_0^{1,p}(\Omega)$ ($C_{n,p}$: $C_1(n,p)$)

$\Rightarrow W_0^{1,p}(\Omega) \subset L^{p^*}(\Omega)$ & this inclusion is bdd (SI-1)

Pf: Step 1: special case $p = 1$, $u \in C_0^1(\Omega)$

Extend $u \equiv 0$ on $\Omega^c \Rightarrow u \in C_0^1(\mathbb{R}^n)$

$u(x_1, x_2, \cdots, x_n) \overset{F.T.C}{=} \int_{-\infty}^{x_i} \frac{\partial u}{\partial x_i}(x_1 \cdots x_i \cdots x_n)\, dx_i$

$|u(x_1, \cdots x_i, \cdots x_n)| \le \int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_i}(x_1 \cdots x_i \cdots x_n)\right| dx_i$, $i = 1, \cdots, n$

$\Rightarrow |u(x_1\cdots x_i\cdots x_n)|^n \le \prod_{i=1}^n \int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_i}(x_1\cdots x_i\cdots x_n)\right| dx_i$

When $p=1$, $p^*=\frac{n}{n-1}$, $\Rightarrow |u(x_1\cdots x_n)|^{\frac{n}{n-1}} \le \left(\prod_{i=1}^n \int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_i}\right| dx_i\right)^{\frac{1}{n-1}}$

$\Rightarrow \|u\|_{L^{p^*}} = \left(\int_{\Omega} |u|^{\frac{n}{n-1}} dx\right)^{\frac{n-1}{n}}$,

$$\int_{-\infty}^{\infty} |u|^{\frac{n}{n-1}} dx_1 \le \left|\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_1}\right| dx_1\right)^{\frac{1}{n-1}} \cdot \int_{-\infty}^{\infty} \prod_{i=2}^n \left(\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_i}\right| dx_i\right)^{\frac{1}{n-1}} dx_1 \quad \text{(Hölder)}$$

$$\le \left(\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_1}\right| dx_1\right)^{\frac{1}{n-1}} \cdot \prod_{i=2}^n \left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_i}\right| dx_i dx_1\right)^{\frac{1}{n-1}}$$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} |u|^{\frac{n}{n-1}} dx_1 dx_2 \le \left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_2}\right| dx_2 dx_1\right)^{\frac{1}{n-1}} \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_1}\right| dx_1\right)^{\frac{1}{n-1}} \cdot \prod_{i=3}^n \left(\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \left|\frac{\partial u}{\partial x_i}\right| dx_i dx_1\right)^{\frac{1}{n-1}} dx_2$$

$$\text{(Hölder)} \le \left(\iint_{\mathbb{R}\mathbb{R}} \left|\frac{\partial u}{\partial x_2}\right| dx_2 dx_1\right)^{\frac{1}{n-1}} \left(\iint_{\mathbb{R}\mathbb{R}} \left|\frac{\partial u}{\partial x_1}\right| dx_1 dx_2\right)^{\frac{1}{n-1}} \prod_{i=3}^n \left(\iiint_{\mathbb{R}\mathbb{R}\mathbb{R}} \left|\frac{\partial u}{\partial x_i}\right| dx_i dx_1 dx_2\right)^{\frac{1}{n-1}}$$

induction: $\int_{\mathbb{R}^n} |u|^{\frac{n}{n-1}} dx \le \prod_{i=1}^n \left(\int_{\mathbb{R}^n} \left|\frac{\partial u}{\partial x_i}\right| dx\right)^{\frac{1}{n-1}}$

$\Rightarrow \|u\|_{L^{p^*}(\mathbb{R}^n)} = \left(\int_{\mathbb{R}^n} |u|^{\frac{n}{n-1}} dx\right)^{\frac{n-1}{n}} \le \prod_{i=1}^n \left(\int_{\mathbb{R}^n} \left|\frac{\partial u}{\partial x_i}\right| dx\right)^{\frac{1}{n}}$ (Basic inequality)

$|\nabla u| = \left(\sum_{i=1}^n \left|\frac{\partial u}{\partial x_i}\right|^2\right)^{\frac{1}{2}} \ge \frac{1}{\sqrt{n}} \sum_{i=1}^n \left|\frac{\partial u}{\partial x_i}\right|$ (By Hölder)

$$\le \sum_{i=1}^n \frac{1}{n} \int_{\mathbb{R}^n} \left|\frac{\partial u}{\partial x_i}\right| dx \le \frac{\sqrt{n}}{n} \int_{\mathbb{R}^n} |\nabla u| dx = \frac{1}{\sqrt{n}} \|\nabla u\|_{L^1(\mathbb{R}^n)} \cdots (*1)$$

Step2: $1<p<n$, replace u by $|u|^\alpha$, $\alpha>1$

$u \in C_0^1(\Omega) \Rightarrow |u|^\alpha \in C_0^1(\Omega)$. $\frac{\partial}{\partial x_i}|u|^\alpha = \alpha |u|^{\alpha-1} u_{x_i} \operatorname{sgn}(u)$. By $(*1)$

$$\||u|^\alpha\|_{L^{\frac{n}{n-1}}} \le \frac{1}{\sqrt{n}} \|\nabla |u|^\alpha\|_{L^1} = \frac{\alpha}{\sqrt{n}} \||u|^{\alpha-1} \nabla u\|_{L^1}$$

$$\le \frac{\alpha}{\sqrt{n}} \|\nabla u\|_{L^p} \||u|^{\alpha-1}\|_{L^q} \quad \text{(By Hölder)}$$

$\frac{1}{p}+\frac{1}{q}=1$, $q=\frac{p}{p-1}$

Take $\alpha>1$ s.t. $\alpha \cdot \frac{n}{n-1} = p^* = \frac{np}{n-p} \Rightarrow \alpha = \frac{p(n-1)}{n-p}$, $(\alpha-1)\cdot q = \left(\frac{p(n-1)}{n-p}-1\right)\frac{p}{p-1} = \frac{np}{n-p} = p^*$

$\alpha \cdot \frac{n}{n-1} = \frac{p(n-1)}{n-p}\cdot\frac{n}{n-1} = \frac{np}{n-p} = p^*$, $\frac{n-1}{n} - \frac{1}{q} = \frac{n-1}{n} - \frac{p-1}{p} = \frac{pn-p-np+n}{np} = \frac{1}{p^*}$

$$\Rightarrow \|u\|_{L^{p^*}} \le \frac{1}{\sqrt{n}} \frac{p(n-1)}{n-p} \|\nabla u\|_{L^p} := C_1(n,p) \|\nabla u\|_{L^p} \cdots (*2)$$

Step3: For general $u \in W_0^{1,p}(\Omega)$

$\exists\, u_k \in C_c^\infty(\Omega)$ s.t. $u_k \to u$ in $W^{1,p}(\Omega)$ $\Rightarrow$ $\begin{cases} u_k \to u \text{ in } L^p(\Omega) \\ \nabla u_k \to \nabla u \text{ in } L^p(\Omega) \end{cases}$

$u_k \to u$ a.e. by passing to a subsequence.

By (*2), $\|u_k\|_{L^{p^*}} \le \frac{\alpha}{\sqrt{n}} \|\nabla u_k\|_{L^p}$. By Fatou Lemma,

$\|u\|_{L^{p^*}} \le \liminf_{k\to\infty} \|u_k\|_{L^{p^*}} \le \liminf_{k\to\infty} \frac{\alpha}{\sqrt{n}} \|\nabla u_k\|_{L^p} = \frac{\alpha}{\sqrt{n}} \|\nabla u\|_{L^p}$

(ii) If $p>n$ & Ω is bdd, then $\forall u \in W_0^{1,p}(\Omega)$, we have $u \in C(\bar\Omega)$, and

$\|u\|_{L^\infty(\Omega)} \le C_2(n,p,|\Omega|) \cdot \|\nabla u\|_{L^p(\Omega)}$ ··· (SI-2) ($\exists v \in C(\bar\Omega)$ s.t. $u \equiv v$ a.e.)

Pf: Assume $|\Omega|=1$. Recall proof of (i), $\forall u \in C_0^1(\Omega)$, $\forall \alpha>1$,

$\||u|^\alpha\|_{L^{\frac{n}{n-1}}} \le \frac{\alpha}{\sqrt{n}} \||u|^{\alpha-1}\|_{L^q(\Omega)} \|\nabla u\|_{L^p}$, $q=\frac{p}{p-1}$ ↓ $\times \left(\frac{\sqrt{n}}{\|\nabla u\|_{L^p}}\right)^\alpha$

$\||\tilde u|^\alpha\|_{L^{\frac{n}{n-1}}} \le \alpha \||\tilde u|^{\alpha-1}\|_{L^q}$ where $\tilde u = \frac{\sqrt{n}u}{\|\nabla u\|_{L^p}}$

$\Rightarrow \|\tilde u\|_{L^{\frac{\alpha n}{n-1}}} \le \alpha^{\frac{1}{\alpha}} \|\tilde u\|^{\frac{1-\frac{1}{\alpha}}{(\alpha-1)q}}_{L^{(\alpha-1)q}}$. Take $\alpha = \delta^m$, $m=1,2,\cdots$, $\delta = \frac{n}{n-1}/\frac{p}{p-1} > 1$

$\Rightarrow \|\tilde u\|_{L^{\delta^m \frac{n}{n-1}}} \le (\delta^m)^{\delta^{-m}} \|\tilde u\|^{1-\delta^{-m}}_{L^{\delta^{m-1}\frac{n}{n-1}}}$, where α

$\Rightarrow \|\tilde u\|_{L^{\delta^m \frac{n}{n-1}}} \le \delta^{\sum_{k=1}^m k\cdot\delta^{-k}} \cdot \|\tilde u\|^{(1-\delta^{-m})(1-\delta^{-(m-1)})\cdots(1-\delta^{-1})}_{L^{\frac{n}{n-1}}}$

$\|\tilde u\|_{L^{\frac{n}{n-1}}} = \frac{\sqrt{n}}{\|\nabla u\|_{L^p}} \|u\|_{L^{\frac{n}{n-1}}} \le \frac{\sqrt{n}}{\|\nabla u\|_{L^p}} \cdot \frac{\sqrt{n}}{n} \cdot \|\nabla u\|_{L^1} \le 1$ (By Hölder)

$\Rightarrow \|\tilde u\|_{L^{\delta^m \frac{n}{n-1}}} \le \delta^{\sum_{k=1}^m k\cdot\delta^{-k}} \le \delta^{\sum_{k=1}^\infty k\cdot\delta^{-k}} = \delta^{\frac{\delta}{(\delta-1)^2}} =: \chi < \infty$. Let $m\to\infty$, $\|\tilde u\|_{L^\infty} \le \chi$

$\Rightarrow \frac{\sqrt{n}}{\|\nabla u\|_{L^p}} \cdot \|u\|_{L^\infty} \le \chi \Rightarrow \|u\|_{L^\infty} \le \frac{\chi}{\sqrt{n}} \|\nabla u\|_{L^p}$ $\forall u \in C_0^1(\Omega)$ ···(*3)

If $|\Omega| \neq 1$, Let $\Omega' = \{y = \frac{x}{|\Omega|^{\frac{1}{n}}} : x \in \Omega\} \Rightarrow |\Omega'|=1$,

$u \in C_0^1(\Omega) \Rightarrow v(y) := u(x)$. $v \in C_0^1(\Omega')$. By (*3), $\|v\|_{L^\infty} \le \frac{\chi}{\sqrt{n}} \|\nabla v\|_{L^p}$

$\|u\|_{L^\infty(\Omega)} = \|v\|_{L^\infty(\Omega')} \le \frac{\chi}{\sqrt{n}} \|\nabla_y v\|_{L^p(\Omega')} \overset{y=x/|\Omega|^{\frac{1}{n}}}{=} \frac{\chi}{\sqrt{n}} \left(\int_\Omega |\nabla u(x)|^p |\Omega|^{\frac{p}{n}} |\Omega|^{-1} dx\right)^{\frac{1}{p}}$

$= \frac{\chi}{\sqrt{n}} |\Omega|^{\frac{1}{n}-\frac{1}{p}} \|\nabla u\|_{L^p}$

For general $u \in W_0^{1,p}(\Omega)$ Take $u_k \in C_0^\infty(\Omega)$ s.t. $u_k \to u$ in $W^{1,p}(\Omega)$

$\Rightarrow \{\nabla u_k\}$ is Cauchy in L^p $\Rightarrow \{u_k\}$ is Cauchy in $C(\bar\Omega)$

$\Rightarrow u_k \to u_\infty$ in $C(\bar\Omega)$ $\Rightarrow u_k \to u_\infty$ in $L^p(\Omega)$ $\Rightarrow u \equiv u_\infty$ a.e. in Ω

$\|u\|_{L^\infty(\Omega)} = \lim_{k\to\infty} \|u_k\|_{C(\bar\Omega)} \le \frac{\alpha}{\pi} |\Omega|^{\frac{1}{n}-\frac{1}{p}} \lim_{k\to\infty} \|\nabla u_k\|_{L^p(\Omega)} = \frac{\alpha}{\pi} |\Omega|^{\frac{1}{n}-\frac{1}{p}} \|\nabla u\|_{L^p(\Omega)}$

Remark: $C_2(n,p,|\Omega|) = \frac{\alpha}{\pi} |\Omega|^{\frac{1}{n}-\frac{1}{p}}$, $\alpha = \delta^{\frac{\delta}{(\delta-1)^2}}$, $\delta = \frac{n}{n-1} / \frac{p}{p-1}$.

$\Rightarrow \delta \to 1$, $\alpha \to \infty$ $(p \to n^+)$ $\delta \to \frac{n}{n-1}$, $C_2 \to$ finite value $(p \to \infty)$

(iii) If Ω is bdd, then $\forall u \in W_0^{1,n}(\Omega)$ & $\forall 1 \le q < \infty$, we have

$\|u\|_{L^q(\Omega)} \le C_3(n,q,|\Omega|) \|\nabla u\|_{L^n(\Omega)}$ (SI-3)

Cor1 (Poincáre's inequality)

$\Omega \subseteq \mathbb{R}^n$ bdd domain. Then $\forall 1 \le p \le \infty$, $\|u\|_{L^p(\Omega)} \le C(n,p,|\Omega|) \|\nabla u\|_{L^p(\Omega)}$ $u \in W_0^{1,p}$

pf: If $1 \le p < n$, $\|u\|_{L^p(\Omega)} \overset{\text{Hölder}}{\le} C \|u\|_{L^{p^*}(\Omega)} \overset{\text{(SI-1)}}{\le} C C_1(n,p) \|\nabla u\|_{L^p(\Omega)}$.

If $p = n$, let $q = n$, $\|u\|_{L^p(\Omega)} \overset{\text{SI-3}}{\le} C_3(n,p,|\Omega|) \|\nabla u\|_{L^p(\Omega)}$

If $p > n$, $\|u\|_{L^p} \le C \|u\|_\infty \overset{\text{SI-2}}{\le} C C_2(n,p,|\Omega|) \|\nabla u\|_{L^p(\Omega)}$

Cor2

$\Omega \subseteq \mathbb{R}^n$ bdd domain. Then $\forall 1 \le p \le \infty$, $W_0^{1,p}(\Omega)$ has an equivalent norm $\|\nabla u\|_{L^p}$

Def1 (continuous imbeded)

X, Y Banach space, $X \subset Y$. X is continuous imbeded into Y if the identity map $i: X \to Y$ is bound i.e. $\|x\|_Y \le C \|x\|_X$ $\forall x \in X$.

note: $X \hookrightarrow Y$

e.g. If Ω bdd, $1 \le p_1 < p_2 \le \infty$, $L^{p_2}(\Omega) \hookrightarrow L^{p_1}(\Omega)$

Cor3 (Sobolev imbedding Thm) $\Omega \subseteq \mathbb{R}^n$ domain, $1 \le p \le \infty$

$W_0^{1,p}(\Omega) \hookrightarrow$
- $L^{np/n-p}(\Omega)$, if $1 \le p < n$ (SI-1)
- $L^q(\Omega)$, if $1 \le q < \infty$, $p = n$, Ω bdd (SI-3)
- $C(\bar\Omega)$, if $p > n$, Ω bdd (SI-2)

strenthen: $C^\alpha(\bar\Omega)$, $\alpha = 1 - \frac{n}{p}$ → Hölder continuous.

Cor4 (General Sobolev Imbedding) $\Omega \subseteq \mathbb{R}^n$, $1 \le p \le \infty$, $k \in \mathbb{N}$.

$$W_0^{k,p}(\Omega) \hookrightarrow \begin{cases} L^{\frac{np}{n-kp}}(\Omega), & \text{if } 1 \le kp < n \\ L^q(\Omega), & \text{if } kp = n,\ 1 \le p < \infty,\ \Omega \text{ bdd} \\ C^\alpha(\bar{\Omega}), & \text{if } kp > n,\ \Omega \text{ bdd}, \text{ where } \alpha = \begin{cases} k - \frac{n}{p}, & p \nmid n \\ k - \frac{n}{p} - 1 + \gamma, & p \mid n,\ \gamma \in (0,1) \end{cases} \end{cases} \quad \cdots (*)$$

Induction + Cor3 $\Rightarrow$ Cor4

$1 \le kp < n$: $u \in W_0^{k,p}(\Omega) \Rightarrow D^\alpha u \in W_0^{1,p}(\Omega)$, $|\alpha| \le k-1 \Rightarrow D^\alpha u \in L^{\frac{np}{n-p}}(\Omega)$, $\forall |\alpha| \le k-1$

$\Rightarrow u \in W_0^{k-1, \frac{np}{n-p}=p_1}(\Omega) \subset W_0^{k-2, \frac{np_1}{n-p_1}}(\Omega) \subset \cdots \subset L^{(p_{k-1})^*}(\Omega)$

$p_{k-1} = \frac{np}{n-(k-1)p}$, $p_{k-1}^* = n \cdot \frac{np}{n-(k-1)p} \Big/ n - (k-1)\frac{np}{n-(k-1)p} = \frac{np}{n-kp}$

Remark: (*) is true if replace $W_0^{k,p}(\Omega)$ by $W^{k,p}(\Omega)$ where Ω is bdd & $\partial\Omega$ is C^1 or even $\partial\Omega \in C^{0,1}$

Def2 (compactly imbedding)

X, Y Banach space. X is compactly imbeded into Y, note: $X \subset\subset Y$ / $X \overset{cpt}{\hookrightarrow} Y$ provided $X \hookrightarrow Y$ and each bdd set in X is precpt in Y. i.e.

K is precpt if $\bar{K}$ is cpt.

Thm (Rellich-Kondrachov Compactness Thm) (RKCT)

$\Omega \subseteq \mathbb{R}^n$ bdd domain. $1 \le p < n$. Then $W_0^{1,p}(\Omega) \overset{cpt}{\hookrightarrow} L^q(\Omega)$ $\forall 1 \le q < \frac{np}{n-p} = p^*$

Remark: $\Omega \subseteq \mathbb{R}^n$ bdd & $p = n$. Then $\forall 1 \le q < \infty$, take small $\varepsilon > 0$ s.t. $q < (n-\varepsilon)^* = \frac{n(n-\varepsilon)}{\varepsilon}$ and thus $W_0^{1,n}(\Omega) \overset{i_1}{\hookrightarrow} W_0^{1,(n-\varepsilon)} \overset{i_2 \text{ cpt}}{\hookrightarrow} L^q(\Omega)$

$\Rightarrow i_2 \circ i_1$ is cpt $\Rightarrow W_0^{1,n}(\Omega) \overset{cpt}{\hookrightarrow} L^q(\Omega)$ $\forall 1 \le q < \infty$

2. $\Omega \subseteq \mathbb{R}^n$ bdd & $p > n$. Then $\forall 0 \le \beta < \alpha = 1 - \frac{n}{p}$, $W_0^{1,p}(\Omega) \overset{cpt}{\hookrightarrow} C^\beta(\bar{\Omega})$

$W_0^{1,p}(\Omega) \overset{cpt}{\hookrightarrow} C^\alpha(\bar{\Omega}) \overset{cpt}{\hookrightarrow} C^\beta(\bar{\Omega})$

3. RKCT + induction $\Rightarrow \Omega \subseteq \mathbb{R}^n$ bdd.

$W_0^{k,p}(\Omega) \overset{cpt}{\hookrightarrow}$ { $L^q(\Omega)$, if $\forall q < \frac{np}{n-kp}$ if $kp \le n$; $C^\alpha(\bar\Omega)$, if $\alpha \in [0, k-\frac{n}{p})$ if $kp > n$. ($\alpha = m+\beta$, $m \in \mathbb{N}$, $\beta \in (0,1)$: $u \in C^m(\bar\Omega)$, $D^\gamma u \in C^\beta(\bar\Omega)$ $|\gamma| = m$)

$W_0^{k,p}(\Omega) \leftrightarrow W^{k,p}(\Omega)$ if $\partial\Omega$ is C^1 or even $C^{0,1}$

Pf of RKCT: Step1: $q=1$ W.T.S: Take any bdd set A in $W_0^{1,p}(\Omega)$, we prove A is precpt in $L^1(\Omega)$ ($\exists M>0$, $\|u\|_{W_0^{1,p}(\Omega)} \le M$ $\forall u \in A$)

Claim: $u \in W_0^{1,p}(\Omega)$, $\tilde u \triangleq$ { u, Ω ; 0, Ω^c } $\Rightarrow u \in W_0^{1,p}(\mathbb{R}^n)$

$A \subset W_0^{1,p}(\mathbb{R}^n)$. Define: $\forall \varepsilon > 0$, $A_\varepsilon = \{u_\varepsilon \mid u \in A\}$, $u_\varepsilon(x) = \int_{\mathbb{R}^n} j_\varepsilon(x-y) u(y)\, dy$

Claim: $\forall \varepsilon > 0$, A_ε is precpt in $L^1(\Omega)$. In fact, $\forall u \in A$,

$|u_\varepsilon(x)| \le \int_{\mathbb{R}^n} j_\varepsilon(x-y)|u(y)|\,dy \le \frac{\|j\|_{L^\infty}}{\varepsilon^n}\|u\|_{L^1} \le \frac{\|j\|_{L^\infty}}{\varepsilon^n} \cdot C\|u\|_{L^p} \le C(\varepsilon, |\Omega|, p) M$.

$\Rightarrow \{u_\varepsilon(x)\}$ uniformly bdd on $\bar\Omega$.

$|\nabla u_\varepsilon| = |(\nabla u)_\varepsilon| = \left|\int_{\mathbb{R}^n} j_\varepsilon(x-y)\nabla u(y)\,dy\right| \le \frac{\|j\|_{L^\infty}}{\varepsilon^n}\|\nabla u\|_{L^1} \le C(\varepsilon, |\Omega|, p)\cdot\|\nabla u\|_{L^p} < CM$

$\Rightarrow \{u_\varepsilon(x)\}$ uniformly equicontinuous on $\bar\Omega$.

By Arzéla-Ascoli Thm, A_ε is precpt in $C(\bar\Omega) \hookrightarrow L^1(\Omega)$

Assume $u \in C_0^\infty(\mathbb{R}^n)$, then $|u_\varepsilon(x) - u(x)| = \left|\int_{\mathbb{R}^n} j_\varepsilon(x-y)(u(y)-u(x))\,dy\right|$

$\overset{z=\frac{x-y}{\varepsilon}}{=} \left|\int_{\mathbb{R}^n} j(z)(u(x-z\varepsilon)-u(x))\,dz\right| = \left|\int_{\mathbb{R}^n} j(z)\int_0^\varepsilon \frac{d}{ds}u(x-sz)\,ds\,dz\right|$ ($|z|\le 1$)

$= \left|\int_{\mathbb{R}^n} j(z)\int_0^\varepsilon \nabla u(x-sz)\cdot(-z)\,ds\,dz\right| \le \int_{\mathbb{R}^n} j(z)\int_0^\varepsilon |\nabla u(x-sz)|\,ds\,dz$

$\Rightarrow \int_{\mathbb{R}^n}|u_\varepsilon(x)-u(x)|\,dx \le \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} j(z)\int_0^\varepsilon |\nabla u(x-sz)|\,ds\,dz\,dx$

$\le \int_0^\varepsilon \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} j(z)|\nabla u(x-sz)|\,dx\,dz\,ds$

$= \|\nabla u\|_{L^1}\int_0^\varepsilon \int_{\mathbb{R}^n} j(z)\,dz\,ds = \varepsilon\|\nabla u\|_{L^1} < \varepsilon C M$.

($u \in C_0^\infty(\Omega) \cap A$).

If $u \in A$, take $u_k \in C_0^\infty(\Omega) \to u$ in $W_0^{1,p}(\Omega)$ hence in $W^{1,p}(\mathbb{R}^n)$

$\|(u_k)_\varepsilon - u_k\|_{L^1(\mathbb{R}^n)} \le \varepsilon \|\nabla u_k\|_{L^1(\mathbb{R}^n)}$

$\downarrow k\to\infty$ $\qquad\qquad$ $\downarrow$

$\|u_\varepsilon - u\|_{L^1(\mathbb{R}^n)} \le \varepsilon \|\nabla u\|_{L^1} \le \varepsilon M\, C(n,p,|\Omega|)$ ···(*1)

4. regularity: $C^\alpha(\overline{\Omega})$ $\quad W_0^{k,p}(\Omega)$ $\quad L^q(\Omega)$ $\quad$ higher $\overset{cpt}{\hookrightarrow}$ lower

$\qquad\qquad\qquad\quad \alpha \qquad k-\frac{n}{p} \qquad -\frac{n}{q}$

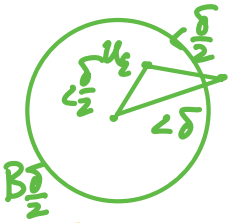
$k-\frac{n}{p}<0$: $k-\frac{n}{p} > -\frac{n}{q} \Rightarrow q < \frac{np}{n-kp}$. $W_0^{k,p}(\Omega) \overset{cpt}{\hookrightarrow} L^q(\Omega)$

$k-\frac{n}{p}=0$: $0 > -\frac{n}{q}$ $\forall q<\infty$, $W_0^{k,p}(\Omega) \overset{cpt}{\hookrightarrow} L^q(\Omega)$

$k-\frac{n}{p}>0$: $k-\frac{n}{p} > \alpha \Rightarrow W_0^{k,p}(\Omega) \overset{cpt}{\hookrightarrow} C^\alpha(\overline{\Omega})$

$\forall \delta>0$, take $\varepsilon = \frac{\delta}{2cM}$. Since A_ε is precpt $\Rightarrow A_\varepsilon \subset$ finitely many balls with radius $\frac{\delta}{2}$ in $L^1(\Omega)$. Thus, $u\in A$, $u_\varepsilon \in$ some ball $B_{\frac{\delta}{2}}$ &

$\|u-u_\varepsilon\|_{L^1} \le \varepsilon \cdot cM < \frac{\delta}{2}$



$u \Rightarrow u\in B_\delta \Rightarrow A \subset$ finitely many balls with radius δ

$\Rightarrow A$ is precpt in $L^1(\Omega)$

Step2: $q\in(1,p^*)$ $\quad \frac{1}{q}\in(\frac{1}{p^*},1) \Rightarrow \exists \lambda\in(0,1),\ \frac{\lambda}{1}+\frac{1-\lambda}{p^*}=\frac{1}{q}$ ···(*2)

Fact: $\Omega\subseteq\mathbb{R}^n$ bdd. $\lambda\in(0,1)$, $1\le p_1\le q\le p_2<\infty$ s.t. $\frac{\lambda}{p_1}+\frac{1-\lambda}{p_2}=\frac{1}{q}$

then $\forall f\in L^{p_2}(\Omega)$. We have $\|f\|_{L^q} \le \|f\|_{L^{p_1}}^{\lambda}\, \|f\|_{L^{p_2}}^{1-\lambda}$

$\forall u\in W_0^{1,p}(\Omega) \overset{SI-1}{\Longrightarrow} u\in L^{p^*}(\Omega)$ +(*2) + Fact $\Rightarrow \|u\|_{L^q} \le \|u\|_{L^1}^{\lambda}\, \|u\|_{L^{p^*}}^{1-\lambda}$ ···(I1)

SI-1: $\|u\|_{L^{p^*}} \le C(p,n)\|\nabla u\|_{L^p} \le C_1 M$, $u\in A$. ···(*3)

Thus: $\|u-v\|_{L^q} \overset{(I')}{\le} \|u-v\|_{L^1}^{\lambda}\, \|u-v\|_{L^{p^*}}^{1-\lambda}$ $\quad u,v\in W_0^{1,p}(\Omega)$

$\le \|u-v\|_{L^1}^{\lambda} (\|u\|_{L^{p^*}}+\|v\|_{L^{p^*}})^{1-\lambda}$

$\overset{(*3)}{\le} \|u-v\|_{L^1}^{\lambda} (2cM)^{1-\lambda}$ ··· (I2)

W.T.S: A is precpt in $L^q(\Omega)$. Take arbitrary $\{u_k\}_{k=1}^{\infty} \subset A$. Since A is precpt in $L^1(\Omega)$. after passing to a subseq, $\{u_k\}$ is Cauchy in $L^1(\Omega)$.

and hence also Cauchy in $L^q(\Omega)$ by (I2) $\Rightarrow A$ is precpt in $L^q(\Omega)$.

L^2-theory for 2nd order elliptic eqs.

Def

Ω bdd domain in $\mathbb{R}^n$, $Lu = -\sum_{i,j=1}^n (a_{ij}(x)u_{x_i})_{x_j} + \sum_{i=1}^n b_i(x)u_{x_i} + c(x)u$, $x \in \Omega$

L is in "divergence form". $\operatorname{div}(A^T \nabla u)$ $A(x) = (a_{ij}(x))_{n\times n}$

Assume $a_{ij}, b_i, c \in L^\infty(\Omega)$

Def

L is strictly elliptic on Ω, if $A(x)$ is symmetric and $\exists \lambda_0 > 0$, s.t.

$A(x) \geq \lambda_0 I_{n\times n}$ a.e. in Ω.

Prob: Existence & Uniqueness of DBVP $\begin{cases} Lu = f & \Omega \\ u = 0 & \partial\Omega \end{cases}$

Def (Weak solution)

- $u \in H_0^1(\Omega) = W_0^{1,2}(\Omega)$ $\forall k \geq 1$, $H_0^k(\Omega) = W_0^{k,2}(\Omega)$ is a Hilbert Space.

  Bound condition. $(u,v)_{H_0^k(\Omega)} = \int_\Omega \sum_{|\alpha|=0}^k \partial^\alpha u \partial^\alpha v \, dx$

- $\forall u \in H_0^1(\Omega)$, Lu is understood as follows:

$$Lu = -\underbrace{(a_{ij}(x)}_{L^\infty}\underbrace{u_{x_i})_{x_j}}_{L^2} + \underbrace{b_i(x)u_{x_i} + c(x)u}_{L^1_{loc}} \to \underbrace{\mathcal{D}'(\Omega)}_{\text{distribution}}$$

- Given $f \in \mathcal{D}'(\Omega)$. $u \in H_0^1(\Omega)$ satisfy $Lu = f$ in Ω $\langle Lu, \varphi\rangle = \langle f, \varphi\rangle$, $\varphi \in \mathcal{D}(\Omega)$

  $\langle Lu, \varphi\rangle = \int_\Omega a_{ij}(x)u_{x_i}\varphi_{x_j} + b_i(x)u_{x_i}\varphi + c(x)u\varphi \, dx$

- For general $f \in \mathcal{D}'(\Omega)$, the DBVP is too hard. → focus on $f \in (H_0^1(\Omega))^* = H^{-1}$

Prob · $Lu \in H^{-1}(\Omega)$? · $H^{-1}(\Omega) \subseteq \mathcal{D}'(\Omega)$?

(1) Yes. $v \in H_0^1(\Omega)$. $\langle Lu, v\rangle_{H^{-1},H_0^1} = \int_\Omega a_{ij}(x)u_{x_i}v_{x_j} + b_i(x)u_{x_i}v + c(x)uv \, dx$

$$|\langle Lu, v\rangle| \le C\|\nabla u\|_{L^2}\|\nabla v\|_{L^2} + C\|\nabla u\|_{L^2}\|v\|_{L^2} + C\|u\|_{L^2}\|v\|_{L^2}$$

$$\le C\|u\|_{H_0^1}\|v\|_{H_0^1}$$

$\Rightarrow Lu : H_0^1 \to \mathbb{R}$ is linear & bdd. hence $Lu \in (H_0^1(\Omega))^* = H^{-1}(\Omega)$ &

$\|Lu\|_{H^{-1}} \le C\|u\|_{H_0^1}$

$\Rightarrow L : H_0^1(\Omega) \to H^{-1}(\Omega)$ is linear & bdd $\|L\| \le C$

(2) Yes. $\forall f \in H^{-1}(\Omega)$. $\langle f, \varphi\rangle \overset{def}{=} \langle f, \varphi\rangle_{H_0^1, H^{-1}}$, $\forall \varphi \in \mathcal{D}(\Omega) \Rightarrow f \in \mathcal{D}'(\Omega)$.

Def (weak solution)

Suppose $f \in H^{-1}(\Omega)$, u is called a weak solution of (DBVP) if

$u \in H_0^1(\Omega)$ & $\forall v \in H_0^1(\Omega)$, $\langle Lu, v\rangle = \langle f, v\rangle_{H^{-1}, H_0^1}$

Remark: (1) weak sol $\Rightarrow$ distribution sol.

(2) If $f \in L^2(\Omega)$, we may regard $f \in H^{-1}(\Omega)$: $\langle f, v\rangle = \int_\Omega fv\,dx$

If $f = g(x) + \sum_{i=1}^n (g_i(x))_{x_i}$, $g, g_i \in L^2(\Omega)$. we may regard $f \in H^{-1}(\Omega)$

$\langle f, v\rangle = \int_\Omega g(x)v(x) - \sum_{i=1}^n g_i(x)v_{x_i}\,dx$

e.g. $\begin{cases} -\Delta u = f & \Omega \\ u = 0 & \partial\Omega \end{cases}$ Then $f \in H^{-1}(\Omega)$, the DBVP $\exists!$ weak sol.

Pf: Weak sol: $u \in H_0^1(\Omega)$ & $\int_\Omega \nabla u \nabla v\,dx = \langle f, v\rangle_{H^{-1}, H_0^1}$ $\forall v \in H_0^1(\Omega)$

$\Downarrow$ Riesze representation Thm

$\int_\Omega \nabla u \nabla v\,dx \to$ inner product?

$(u, v)_{H_0^1(\Omega)} = \int_\Omega uv\,dx + \int_\Omega \nabla u \nabla v\,dx$ By Poincaré Ineq $\Rightarrow \|u\|_{H_0^1} = \|u\|_{L^2}$

$(u, v) = \int_\Omega \nabla u \nabla v\,dx \to$ inner product.

$\forall f \in H^{-1}(\Omega)$. $\exists! u \in H_0^1(\Omega)$ s.t. $\int_\Omega \nabla u \nabla v\,dx = \langle f, v\rangle_{H^{-1}, H_0^1}$ $\forall v \in H_0^1(\Omega)$

Thm (Lax-Milgram) generalization of Riesz Rep Thm.

Let H be Hilbert space and $a(\cdot,\cdot)$ be a bilinear form on H.

Assume $\exists$ Constant $M, \gamma > 0$ s.t. $\begin{cases} |a(u,v)| \le M \|u\| \|v\| & \forall u, v \in H \\ \gamma \|u\|^2 \le a(u,u) & \forall u \in H \end{cases}$

Then $f \in H^*$, $\exists!\ u \in H$ s.t. $a(u,v) = \langle f, v \rangle_{H^*, H}$ $\forall v \in H$.

$$\|u\|_H \le \frac{1}{\gamma} \|f\|_{H^*}$$